

RoboMaster Competition

Game Design Document · Worked Sample

SAMPLE: this is a fully worked example so you can see what “finished” looks like. Do not submit this game. Pick or invent your own; first crew to claim an idea keeps it.

Crew Information

Crew name / number:

Crew 5 (sample)

Section:

High School, Comp Sci/Robotics 01T1

Date:

Draft: Oct 20, 2026 · Final: Nov 10, 2026

Crew roster and lead roles

Role	Crew Member	Contact
Pilot	Student A	in class
Programmer	Student B	in class
Engineer	Student C	in class
Media and Rules	Student D	in class

Required Fields

1. Game name and the one-line pitch

The Blackbeard Blockade.

One robot runs the reef blind, on autopilot, then the pilot takes the wheel and has forty-five seconds to put three cannon shots through the blockade before the tide clock runs out.

2. Setup

A 12 x 16 foot field taped on the gym floor. A “reef” lane down the middle, marked with five vision markers spaced about two feet apart in a gentle zig-zag, is the autonomous section. Past the reef, three cardboard “blockade” targets (numbered 1 to 3) stand in a row eight feet from the finish line, each about the height of the robot's blaster.

Built from gym tape, cones for the reef edges, and cardboard targets with the numbers taped on. Field sets up in under ten minutes, which is an Engineer deliverable, not just a nice-to-have.

3. How to play

One robot per run, one crew at a time. The match clock is 90 seconds total: up to 45 seconds for the autonomous reef run, then the pilot has whatever time is left (minimum 45 seconds guaranteed even if the auto run was fast) to drive up and fire on the blockade.

The clock starts the moment the referee says “shove off” and the program is started. It stops when the pilot calls “done” or time runs out, whichever comes first.

4. How you score and how you win

Reef phase: 5 points for every vision marker the robot reads correctly in order, hands off (up to 25). Touching a reef cone costs 5 points. Leaving the reef lane entirely ends the autonomous phase early with whatever points were already earned.

Cannon phase: 10 points per blockade target hit, in any order (up to 30). A hit is confirmed by the target's armor panel registering, not by eye.

High score wins. A tie is broken by whichever crew finished with more time left on the clock; if that also ties, both crews re-run the cannon phase only, sudden death, first hit wins.

5. Rules and penalties

Eye protection on for every run, no exceptions, same as class safety rules. Beads are used for the cannon phase, so anyone within ten feet of the field who is not wearing eye protection has to step back before the pilot is allowed to fire.

The robot may not be touched during the reef phase. A crew member touching the controls during autonomous ends that phase immediately at the current score, zero bonus.

Driving through a blockade target instead of shooting it does not count as a hit and costs 5 points for unsafe driving near a person-height obstacle.

A robot that leaves the taped field boundary at any point is stopped by the referee and the run ends at the current score.

6. Referee script

1. Confirm eye protection on everyone within ten feet of the field, confirm the blaster is loaded correctly for the beads phase only.
2. Ask the crew “ready?”, start the clock and say “shove off” the moment the crew starts the autonomous program.
3. Call out each marker read aloud as it happens (“marker one, marker two...”) so the crowd can follow along.
4. The instant the auto run ends (marker five read, boundary left, or 45 seconds up), call “hands on, cannon phase” and hand the controller to the pilot.
5. Call each confirmed hit out loud as it registers. Call time at zero or when the pilot calls “done.”
6. Announce the final score to both crews and the crowd before resetting the field for the next run.

Bonus Fields

Bonus: Field diagram

See attached sheet: 12 x 16 ft field, reef lane centered and 3 ft wide with five markers on a zig-zag spaced 2 ft apart, blockade targets in a row 8 ft past the reef exit, 4 ft apart, start line marked at the near edge.

Bonus: Program notes

Block program in the RoboMaster app's Lab mode. Reads each vision marker in sequence using the marker-recognition block, drives forward a fixed distance between markers (tuned by trial runs), and stops automatically if no marker is found within 3 seconds (safety timeout so the robot never just wanders off the field). A variable tracks how many markers have been read so the crew can display a running count.